Adaptive CBRAM Synapse Model for Emerging Biohybrid Computing Systems

G. Kleitsiotis[1], I.K. Chatzipschalis[1,2], T. P. Chatzinikolaou[1], I. Tompris[1], I.-A. Fyrigos[1], A. Adamatzky[3], P. Ayres[4], I. Vourkas[1], G. Ch. Sirakoulis[1]

1. Democritus University of Thrace
2. Universitat Polit'ecnica de Catalunya
3. University of the West of England
4. Royal Danish Academy

Overview

Motivation:

- **Challenge: Biohybrid computing [1] requires *both* fast spiking and plasticity**
- **Diffusive memristors: Volatile, unipolar RESET – Can emulate neural nodes [3]**
- **Adaptive CBRAMs: Non-volatile, bipolar, adaptive switching – Synaptic counterparts [2]**

Our Proposal:

- **Build a hybrid neuromorphic pathway where:**
  1. **A volatile memristor within a LIF neuron's soma generating action potentials**
  2. **An adaptive CBRAM acting as a synapse element**

Adaptive CBRAM Modelling

➢ **Filament Evolution →**

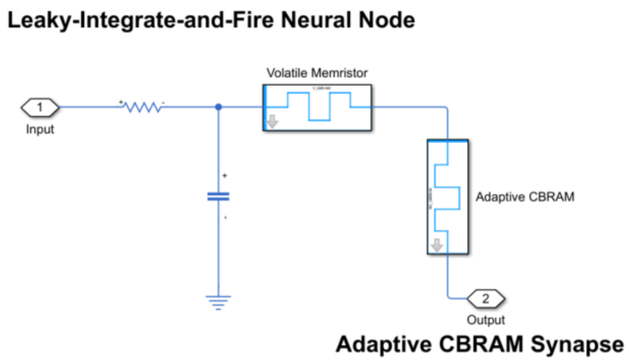
1. **Drift + f(D) * Diffusion + g(D)* Thermal**

2. $$\frac{dD}{dt} = \frac{{}^{V}\max\, {}^{E}\text{abs}}{K_m + {}^{E}\text{abs}} - \eta D \,,$$

Michelis – Menten Kinetics

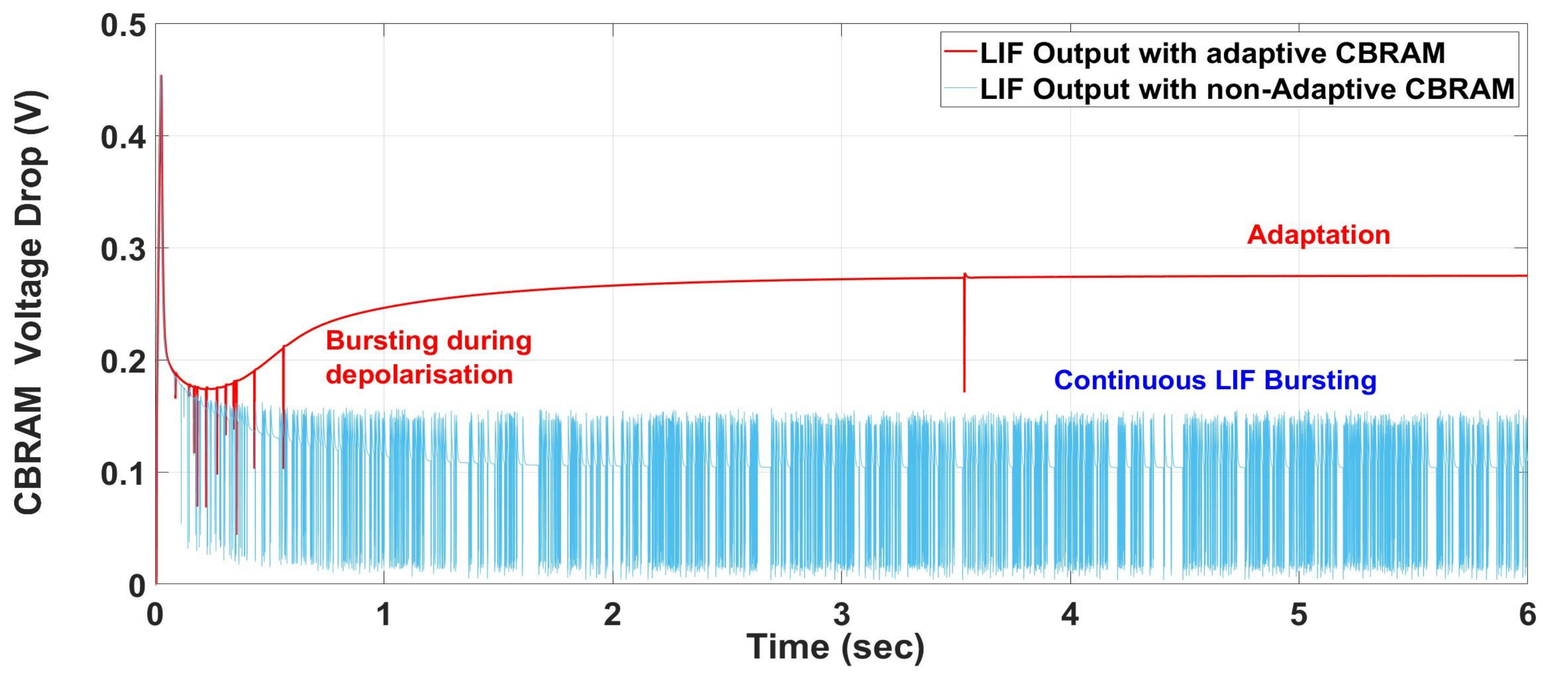
The model is based on a standard Drift-Diffusion-Thermal Diffusion Multiphysics system [2], Augmented with an Adaptation mechanism on the diffusion and thermal terms, derived by Michaelis-Menten kinetics [4]

Bioinspired Neural Pathway



- **Volatile Memristor provides LIF neuron dynamics (fast spiking thresholding, reset)**
- **CBRAM introduces long-term adaptation due to its unique filament dynamics**
- **Results in a physically consistent neuromorphic primitive**

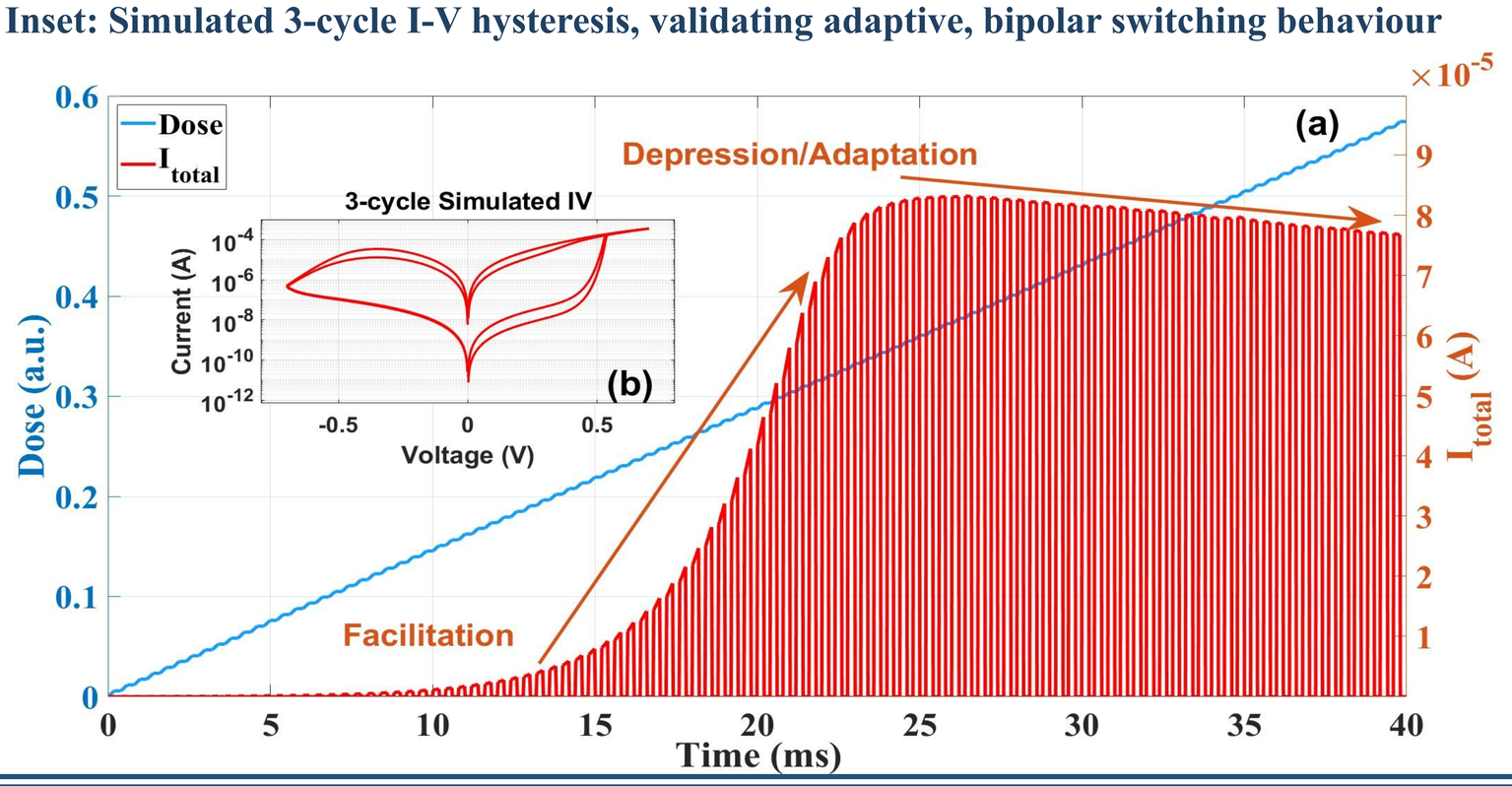
LIF Behaviour Comparison with and without adaptive CBRAM Terminal



- **Pure LIF (No Synaptic terminal): Continuous bursting under sustained input. No regulation as time progresses**
- **Adding the Adaptive CBRAM terminal, we can observe:**
  - ✓ **Early pulsed behaviour → Bursting during initial depolarisation**
  - ✓ **Dosage accumulation → Adaptive conductance supresses runaway activity**
  - ✓ **Network Stabilisation → Exhibiting spike frequency modulation – controlled bursting**
- **The adaptive CBRAM converts a volatile neural node into a self-regulating spiking element with embedded plasticity, realistically recreating the dynamics observed in its biological counterparts**

Adaptive CBRAM Synaptic Response – Dosage Accumulation

- **The simulated CBRAM pathway exhibits short term facilitation followed by long-term depression/adaptation under repeated pulsed stimulation.**
- **The blue curve shows the accumulated dose (absorbed energy), while the red trace shows the total current response.**
- **As the energy dosage accumulates, the plasticity trace saturates and the current response decreases (adaptation)**
- **Inset: Simulated 3-cycle I-V hysteresis, validating adaptive, bipolar switching behaviour**



Future Work and Conclusions

- **We introduced a compact, physics-based model for adaptive CBRAM synapses, governed by an interdependent set of equations for drift, diffusion and thermodiffusion. We have also added an energy-dose driven plasticity mechanism, enabling long-term resistivity changes with non-linear saturation**
- **We have proven that when combined with a volatile memristor-based LIF node, the system converts input frequency and amplitude into adaptive conductance states**
- ***The model thus provides a consistent computational primitive where memory, adaptation, and computation emerge from device interactions!***
- **In the future we intend to scale this primitive to larger neuromorphic networks and explore mapping to biohybrid platforms (fungi/mycelia) as adaptive computing substrates.**

References

[1] A. Adamatzky *et al., Logica Universalis,* 16(4), 655, 2022

[2] G. Kleitsiotis, *et al., IEEE Trans. Electron Devices,* 70(6), 3322, 2023

[3] Z. Wang, *et al., Nat. Mater.,* 15, 2016

[4] L. Michaelis and M. Menten, *Biochem. Z.,* 333-369, 1913

Acknowledgements

This work has been partially supported by the framework of the FUNGATERIA project, which has received funding from the European Union's HORIZON-EIC-2021-PATHFINDER CHALLENGES program under grant agreement No. 101071145.